

TQM MAGAZINE - VOL.1

MAGAZINE FROM TQM DEPT
JSW SEEL LTD.
VIJAYANAGAR WORKS
SEP 15, 2023
FRIDAY



QUALITY CIRCLE

蟬
改善

TO ACHIEVE ZERO - INJURY,
DEFECT, WASTE, BREAKDOWN

*FOR INTERNAL
CIRCULATION ONLY



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EVOLUTION OF QC

1962

First QC Circle was registered with QC Circle Head Quarters in Japan.



1977

International Association of Quality Circles (IAQC) was formed in USA.



1974

Lockheed Company, USA started Quality Circle movement.

1980

BHEL, Hyderabad first in India to start Quality Circles



1982

Quality Circle Forum of India (QCFI) was founded.

2008-2012

- Familiarizing the employees with the concept of Quality Circles and creating awareness about its benefits
- Creating middle management appreciation and support
- Conducting rigorous training programs on TQM tools

2012-2018

- Deriving operational benefits from the usage of TQM tools
- Benchmarking TQM systems and creating organization wide support
- Designing initiatives and organization wide campaigns
- Creating shared practices and codification of practices

2018-2023

- Lean Safety Circle initiated
- Jury development program
- Participation of employees in Jury panel

2023-2025

- Development of QC portal for project reviews
- Reviewing the TQM practices
- Diversifying themes of Quality Circle

EVOLUTION OF QC IN JSW



What are QC Circles?

A QC Circle is a small group consisting of front-line employees who continually control and improve the quality of their work, products, and services.

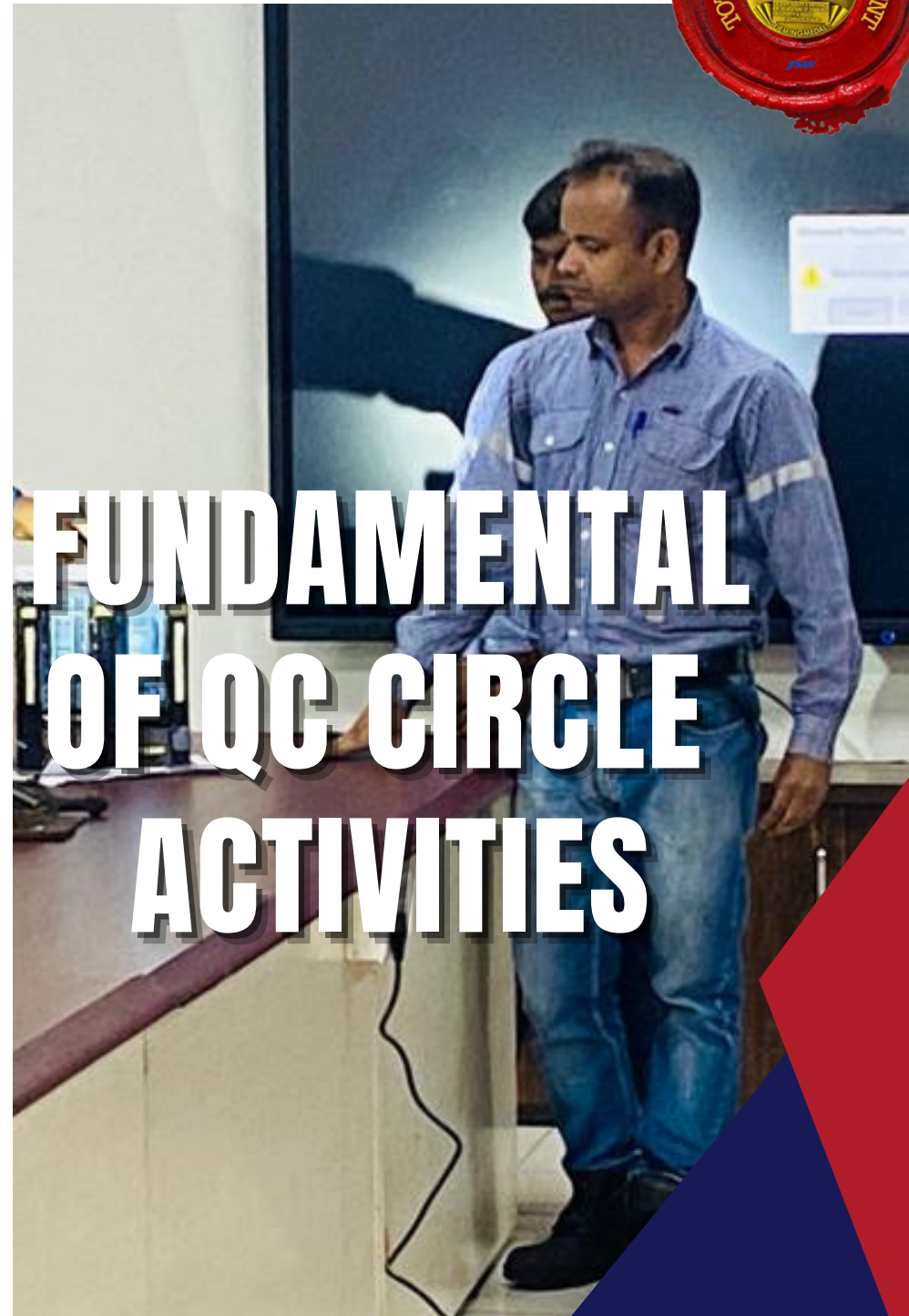
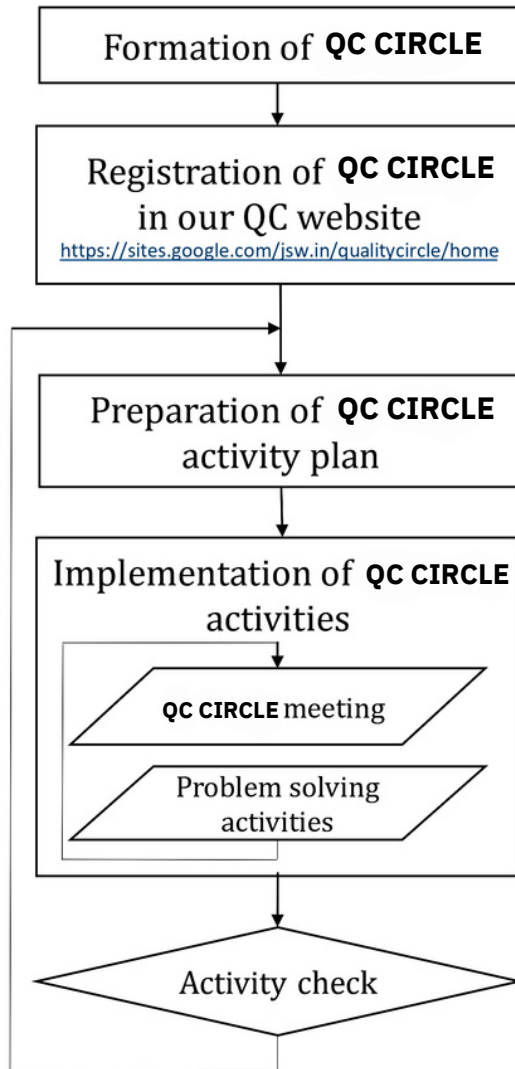
These small groups operate autonomously, utilize quality control concepts and techniques and other improvement tools tap members' creativity, and promote self and mutual development. It is a voluntary activity.

QC Circle activities aim to: develop members' capabilities and achieve self-actualization, make the workplace more pleasant, vital, and satisfying, improve customer satisfaction, and contribute to society.

If you are now aware of the benefits of QC Circle, here's a guide for you on how to proceed with QC Circle activities. Form a QC Circle by including employees working in the same department who volunteer to solve problems and improve processes within their department. Then it is mandatory to register your QC Circle on our QC website. After identification of problems create an activity plan outlining your goals, and objectives and form a roadmap. Now to implement what you planned for, you need to hold regular meetings to discuss progress, share ideas, foresee any possible resistance, and collaborate on problem-solving. These meetings will provide a platform for communication and coordination within the group.

After implementation, you need to evaluate the results and check if the desired achievements are achieved or not.

If it is "YES" congratulations, now you may proceed with the next activities (including the standardization of the new process), but if not then you may need to revisit the QC circle activity plan and make the necessary adjustments.





A leader is a person chosen by the members of the Circle on consensus. He is the most important person in a Quality Circle. For him to play an effective role, he should be given the requisite education and training to lead and to develop leadership qualities and skills. Since Quality Circle conceptually is also 'applied management' at grass root level, they are to be given training on all those aspects.

Aside from the Facilitators, the success of the Quality Circle will depend upon how well the leader leads the group. The group, whose members have come voluntarily to form the group, chooses him on consensus. Any member of the Quality Circle can become the leader of it. Normally on a consensus basis, a leader is decided.

A QC leader must remember to work as a team.

- Voluntary members are from the same work area and know one another well.
- Members are to be closely knit to work as a team.
- For group work to be meaningful, members should interact with one another adequately.
- Members should share common objectives.
- There should be openness in the group.
- Members should respect one another.
- All should be equal partners in participation.
- The independence and initiative of each member should be protected.
- Information available should be shared without restrictions.

- A QC leader has to lead the group but should not dominate the proceedings.

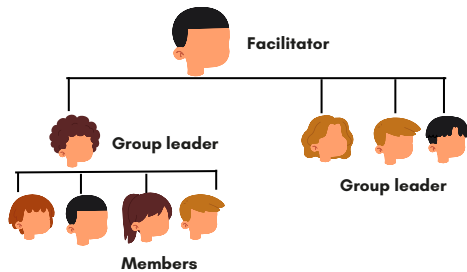
Role and Functions of a QC Leader

Unlike a regular work leader, A QC leader plays a much more difficult role. QC working needs self-discipline. A QC leader should be a highly disciplined person. He/ She should conduct meetings regularly. He/she should know group dynamics and interactive behaviors for purposeful group function. He/She is expected to be task-oriented and at the same time play a socio-emotional part too. He/She is the one who should lead his/her team towards goals. He/She should ensure that records of circle activities are maintained properly. He/She should work to plan and assign roles to members and prepare the team for management presentation and ensure that everyone is involved in one way or other, understanding their capability and assigning work accordingly. He/she needs to be knowledgeable and should be able to clarify the doubts of group members. If he/she does not know the answer, he/she should find out from the right person and convey it to them. He/She should not take credit for himself/herself but give credit to his/her group members when the group wins. He/she should send a monthly Quality Circle activity report to all concerned.





Quality Circle Member is one among the equals of a group. He/She becomes a member of a Quality Circle out of his/her own volition and with an understanding of the concepts and philosophy. If he/she is interested in his/her growth and development, he/she should get involved wholeheartedly.



How do Members join Quality Circle?

After exposure to the Quality Circle concept, employees at the grass root level join QCs voluntarily, after fully understanding the concept and philosophy of the Quality Circle. Many feel that there is no need to have it as a voluntary movement. The fear is unfounded and if people do not volunteer it is because they do not have proper knowledge about the concept. If this concept is institutionalized keeping the basic aim of self and mutual development of the employees only a very few do not volunteer.

Role and functions of Members

Unlike any other concept or form of small group working, the leadership in Quality Circle is rotatable.

Today's Members could be tomorrow's Leaders. Therefore, it is essential that each member is involved and develops himself/herself to take up a leadership role at a later date. As a QC member, he or she should:

- Understand why QC group is formed and why one is a member.
- Attend QC meetings regularly.
- Cooperate with the leader and perform roles assigned to him/her efficiently.
- Respect other members and cooperate with them.
- Learn to listen, appreciate, accommodate, and assimilate others' viewpoints or opinions.
- Share QC teams' goals and objectives.
- Interact with members freely.
- Do the homework before every meeting.
- Contribute to the success of the group.

ROLE AND FUNCTIONS OF A QC MEMBER





Give support to QC Circle activities and be prepared to lead them when needed. Your backing must be based on your true understanding of the fundamentals of QC Circle activities as well as of actual conditions within your organization. To know QC Circle, you must attend QC Circle conferences and other related meetings both within and outside your own company. Learn from the activities of other companies.

Remember that QC Circle activities are voluntary activities. Do not voice your opinion to the interface, and let the activities move at their own pace. Trust the people who are working for you. Proceed with the notion that man is by nature good.

QC Circle activities are conducted because of respect for humanity. They have as their goals the enhancement of people's abilities. This will, in turn, help the individuals involved, the divisions, and the company. QC Circle can never be conducted just for the benefit of the company alone.

As long as there is a workplace, QC Circle activities must be continued. They are not a fad for this particular moment.

Show your support for QC Circles not by words but by action. Help them when they are establishing their goals when they are setting up their organizational structures (for example, scheduling meetings for leaders and promoters) when they are making educational plans; when they are organizing QC seminars or conferences (within their divisions, etc.), and when they are sending people outside the company.

Help them to establish guides, concrete plans and help them in their implementation.

QC Circle must meet at least twice a month, preferably once a week. A circle that meets only Once a month is a sleeping circle. Some of your workers may want to have more frequent meetings. If you are their supervisor, do not say, "We are too busy now, don't have a meeting this week."

The busier the workplace, the more appropriate it is to have a QC Circle meeting. That circle must investigate the reason why everyone must work so hard. A fundamental solution is needed. QC Circle activities are inseparable from the company's day-to-day work. Some people consider QC Circle to be an additional burden. This attitude is of course wrong, and supervisors must find a way to correct it.

Do not seek immediate results, study first. Foremen and workers will become better trained through the circle activities and before long results will become noticeable. As managers and supervisors, you must be patient. Take time to nurture circle activities and promote them.

Supervisors have a lot to contribute. They can help the circles get started, approve the themes they propose, and can check their activity plans and reports. They can also help them find a time and place to meet. They can help assemble and prepare data and materials. They can provide for QC activities perquisites when needed. They can help make awards and create channels through which suggestions of QCCs can be accepted.

DOS AND DON'TS FOR MANAGERS AS WELL AS FOR SUPERVISORS BY DR. K. ISHIKAWA



Our organization conducts **intradepartmental and interdepartmental QC competitions** amongst all Quality Circles, which provide cross learning platform.

Winning teams get the opportunity to participate in external competitions organized by QCFI (Quality Circle Forum of India) and the Confederation of Indian Industries (CII), etc.

Evaluation Criteria

The evaluation criteria during external and internal competition are based on the objective of the development of grassroots employees. It consists of an evaluation of the QC record book, a knowledge test based on problem-solving tools and techniques, an evaluation of the case study, a case study presentation, and a question-and-answer session after the case study presentation. Questions are based on the case study preparation to judge the knowledge and involvement of each member on the project presented.

Quality Control Circles create a conducive and stimulating atmosphere for the employees such that their creativity can be energized and properly canalized. QCC competition provides a unique opportunity to motivate employees to form Quality Circles within the company and also provides a unique opportunity to motivate employees to form Quality Circles within the company and also provides them a forum to exchange ideas and share their experiences. It is beneficial for senior managers, supervisors, and management personnel.

Purpose of QC competition

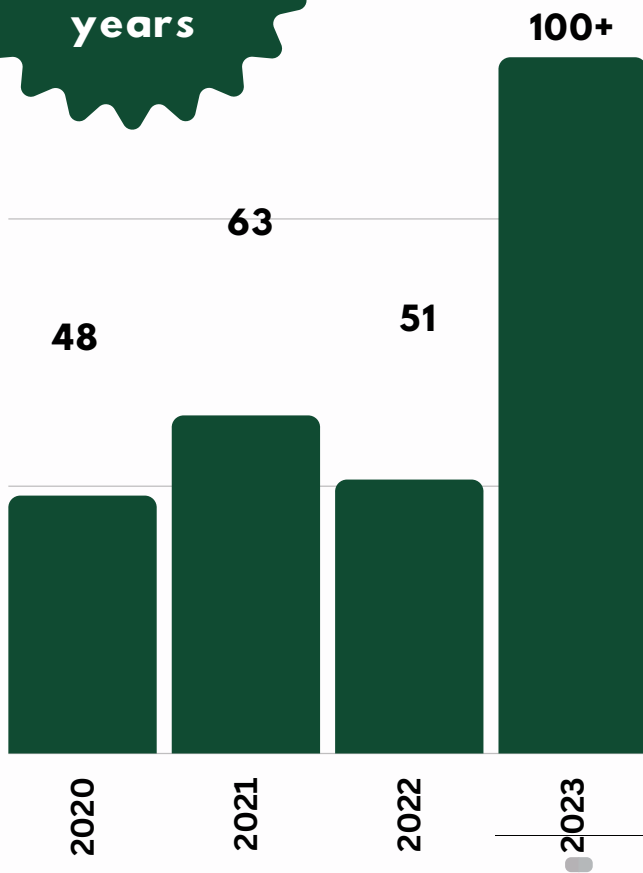
- Enables you to benchmark abilities
- Knowledge sharing for building competencies
- Enhancing creativity and critical thinking
- Recognizing and rewarding exemplary work
- Deepens their knowledge about the subject of interest through exemplification.
- Having healthy competition boosts your confidence to face the crowd
- Disseminating knowledge and ideas
- Provides a unique platform to develop a sound relationship with other organizations
- Helps in identifying strengths and weaknesses of teams and individuals
- Helps in developing leadership qualities



GLIMPSES OF QC PRESENTATIONS 2023



Highest QC
participation
in past 4
years





From August 1st to August 12th, over 100 teams participated in QC Presentations, showcasing their projects and innovations.



Do you know! **The most asked question by the panelists** to presenters was, why do we do Quality Circles and who benefited the most from it?





In response, the participants offered an array of answers that showcased their diversity of thought. Yet, amidst these varied explanations, there still might be an essential element that is yet to be brought to light, calling for further exploration and fruitful discussion.



A Quality Circle is a **group of employees**, who regularly meet to identify, analyze, and propose solutions to workplace challenges. This **voluntary** activity aims to improve work processes, enhance product quality, and foster team collaboration.



JSW Steel Limited

Delivering Growth in Challenging Times

Along with QC Circles, we have Allied QC Circles, which is a continuous improvement activity consisting of various methodologies. Concepts other than the infamous Quality Circle such as **Lean Six Sigma, Lean Quality Circle, 5S, etc.** are called allied concepts.



QC Circle is a unique blend of employees from different hierarchies combined together **against a problem.**

QUALITY CIRCLE PROJECT EVALUATION CRITERIA



Quality Circle Project Evaluation Criteria

Steps	QC Steps	Categories	Marks	Max. Marks
1	Identification of problems	Up to 19	1	5
		20-29	2	
		30-39	3	
		40-49	4	
		50 & Above	5	
2	Selection of problem	A,B,C Categorization	1	5
		Priority / Ranking Method	2	
		Base used for priority	2	
3	Define the problem	Milestone chart	1	10
		Flow diagram/pictorial diagram of the process	2	
		Description of the problems with labelled diagram or any other creative way	5	
		Objectives (Minimum 4)	1	
		Goals/ Target	1	
4	Analysis of the problem	Data / verbal data analysis by 4W-1H	2	10
		Impact on the performance parameters like P,Q,C,D,S,M,E	2	
		Stratification of data pareto chart, Histogram, Control Chart, Stratification etc. as appropriate applicable	6	

Make sure to align your presentation with the evaluation criteria to grab maximum points!

QUALITY CIRCLE PROJECT EVALUATION CRITERIA



Quality Circle Project Evaluation criteria				
Steps	QC Steps	Categories	Marks	Max. Marks
5	Identification of problem	Fish Bone diagram with proper headers	2	10
		Cause , Sub-cause, Sub-Cause etc.	5	
		Marking/ Listing probable causes	3	
6	Finding out the root causes	Use of appropriate method for causes validation	5	5
7	Data Analysis	Collection of appropriate data for the validation of root causes	5	10
		Analysis of data using proper tool and techniques	5	
8	Development of solution	Minimum 05 number of solution to be generated using brainstorming	5	10
		Selection of best possible solution	5	
9	Foreseeing probable resistance	Minimum 05 probable resistance to be identified along with the solution to the identified resistances	5	5
10	Trial implementation and check performance	Use of PDCA	5	15
		trial implementation	5	
		Check performance	5	
11	Regular implementation	Current Status	5	10
		Over all gain: Tangible and Intangible	5	
12	Follow up review	Follow system developed, revised SOP's SMP's	5	10
		Review results, Monitoring System	5	

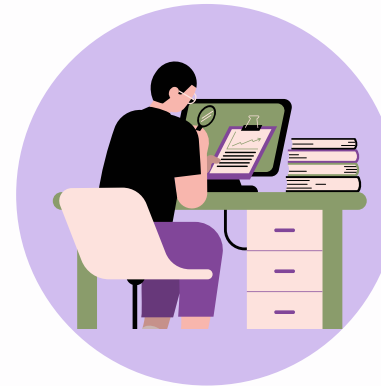


ELIGIBILITY TO FORM A QC CIRCLE

Member	• Work face to face • Encourage each other, and affect each other • Get to know each other well and be able to talk freely • Share common purposes
Leader	Facilitator, Leader, or a Member mutually elected
No. of people	5-10 are desired
Name of the QC Circle	A professional name, that can express the willingness of the QC Circle activities, should be decided on by all the members

4 PRINCIPLES OF BRAINSTORMING

NEVER CRITICIZE



The person who was criticized will not speak up

MANY ARE WELCOME



Many options and ideas are welcome. Quality can bring forth quality

FREE IDEAS



Free ideas generate startling or novel options

COMBINING IDEAS



Follow suit, and combine the opinions of others



Stratification

Stratification is the process of dividing members of a data population into homogeneous subgroups before sampling according to common factors or tendencies. It is a technique used to segregate the data from a variety of sources to obtain the trends.

Check Sheets

Check Sheets are simple and effective tools used to collect and record data in a structured manner. They help in categorizing and analyzing data to identify patterns or trends. By using check sheets, Organizations can easily track and monitor various parameters, such as defects, errors, or customer complaints, allowing them to make informed decisions based on accurate data.

Histogram

Histograms are graphical representations of data distribution. They provide a visual depiction of the frequency and distribution of a set of data. By using histograms, organizations can identify patterns, variations, or abnormalities in data, enabling them to focus on areas that require improvement or attention.

Pareto Charts

Pareto Charts are bar charts that prioritize problems or causes based on their frequency or impact. They follow the Pareto principle, also known as the 80/20 rule, which states that roughly 80% of the problems are caused by 20% of the causes. By using Pareto charts, organizations can focus their efforts on the vital few causes that contribute the most to the overall problem.

BASIC QC TOOLS

A systematic approach to problem solving



Cause and Effect Diagrams

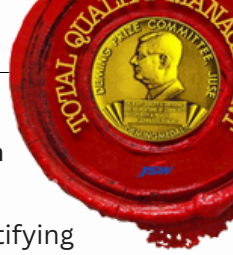
Cause and Effect Diagrams (also known as Fishbone or Ishikawa diagrams) are graphical tools used to identify and analyze the potential causes of a problem. By categorizing causes into different branches, these diagrams enable organizations to systematically investigate and understand the root causes behind a particular problem/effect.

Scatter Diagrams

Scatter Diagrams are used to identify and analyze the relationship between two variables. They help in understanding the correlation between different factors, allowing organizations to make data-driven decisions. By plotting data points on a graph, organizations can visualize the relationship and determine if any changes in one variable affect the other.

Control charts

Control Charts are statistical tools used to monitor and control process variations over time. They provide a graphical representation of process performance, allowing organizations to identify if a process is within control limits or experiencing any unusual variations. By using control charts, organizations can take corrective actions to maintain stable and predictable processes.



Brainstorming

A group creativity technique designed to generate a large number of ideas for the solution to a problem. It encourages free thinking and open discussion within a group to explore all potential ideas without immediate criticism.

Affinity Diagram

A tool used to organize and cluster large amounts of data or ideas into naturally related groups. It's particularly useful when you have lots of raw data or ideas and need to group them into themes or categories.

Arrow Diagram

Also known as an activity network diagram, it depicts the sequence of activities needed to complete a project and the dependencies between them. It allows teams to determine the critical path and potential bottlenecks.

Tree Diagram

A systematic approach to displaying the hierarchical nature of a structure in a graphical format. It breaks down broad categories into finer levels of detail, helping to understand the relationship between different layers.

ADVANCE QC TOOLS

A systematic approach to problem solving



Process Decision Program Chart (PDPC)

A tool used for identifying potential risks in achieving objectives. It systematically identifies what might go wrong in a plan under development, aiding in preparation and mitigation.

Matrix Diagram

A visual representation tool that illustrates the connection or correlation between two or more sets of information. It aids in determining relationships between factors, causes, and objectives.

Matrix Data Analysis

A complex mathematical technique used to analyse the relationships between multiple variables. It helps in understanding patterns, correlations, and dependencies among multiple data sets.

Relation Diagram

Also known as an interrelationship diagram, it helps identify and analyse the causal relationships among a set of ideas or issues. It is particularly useful in complex scenarios where multiple causes and effects are interlinked.



S NO.	DIAGNOSTIC QUESTION	RESPONSE
1	Is there an intractable problem?	
2	Is the work tiring hard to do?	
3	Is there waste, strain or lack of uniformity?	
4	Are there any items that are defective or need doing over?	
5	Are there any requests of complaints from upstream or downstream processes or from other workplaces?	
6	Is there anything that inconveniences customers?	
7	Is the work done accurately and quickly?	
8	Are you achieving more results with less manpower?	
9	What are the policies and targets of the squad or section?	
10	Is there anything that you particularly want to accomplish?	

10 WAYS TO IDENTIFY PROBLEMS

GEMBA WALK

- Pick a theme such as productivity, cost efficiency, safety, etc
- Prepare your team
- Focus on the process, not on people to observe, understand, and improve the process
- Be where the value stream is to identify areas with a high potential for waste activities
- Record your observations and don't make suggestions during the walk
- Have an extra pair of eyes
- Follow up with the team on what you have seen

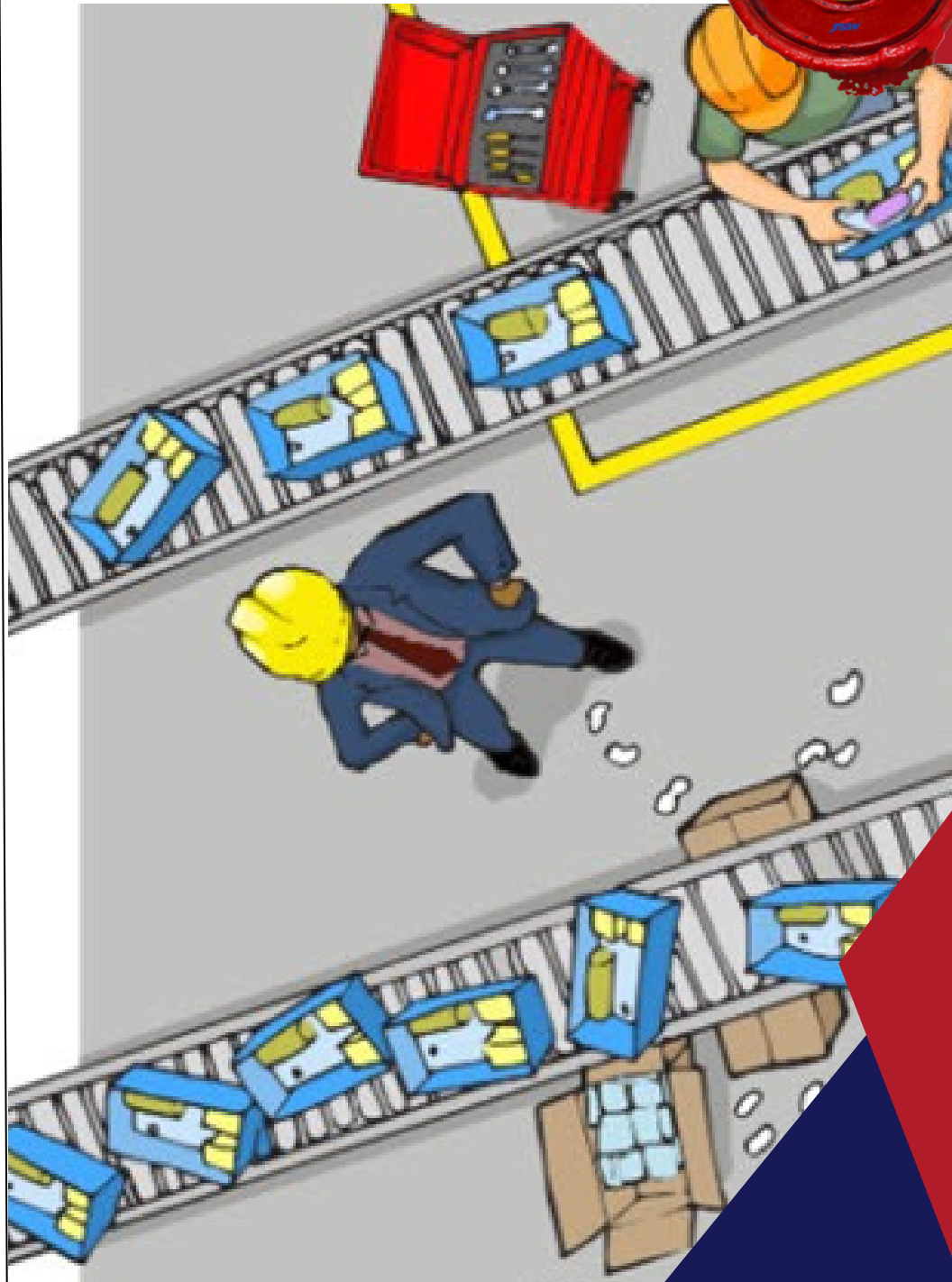
How can you be present everywhere to find out problems and provide solutions at the same time?

To improve overall efficiency, you need to spot opportunities and solve problems in your process. But how you can be present everywhere to find out problems and provide solutions at the same time? So the next question comes,

Is it possible to improve the process and find opportunities for improvement in your department?

The answer is yes: a simple but effective Gemba walk. Gemba Walk is an idea that aims to reduce waste or defects to reduce inefficiencies within the process. It also aims to improve safety practices and procedures by observing the actions required to complete daily tasks and identifying ways to make work safer. It is a Japanese word that translates to “the actual place” which means that managers get out of their offices and visit workers in their environment to look for opportunities and to improve the process which can make them achieve their production outputs effectively and efficiently. The goal of Gemba Walk is to become more involved in day-to-day operations that you normally don't see which helps to make more realistic and informed decisions.

**In this picture: Illustration of Gemba walk*





Quality circles have long been considered as a powerful tool for improving productivity, enhancing employee engagement, and fostering a culture of continuous improvement within organizations.

However, it is evidenced that “extrinsic motivation” has set in our ecosystem generating negative vibes. It seems that in many cases, the primary focus of quality circles has shifted from their original purpose to simply winning awards and gaining external recognition. Whereas the TQM is all about “intrinsic motivation”.

One cannot deny the allure of awards and recognition. They serve as a validation of hard work, dedication, and success. However, when these accolades become the sole objective of quality circles, the true purpose of these initiatives gets lost in the pursuit of external validation.

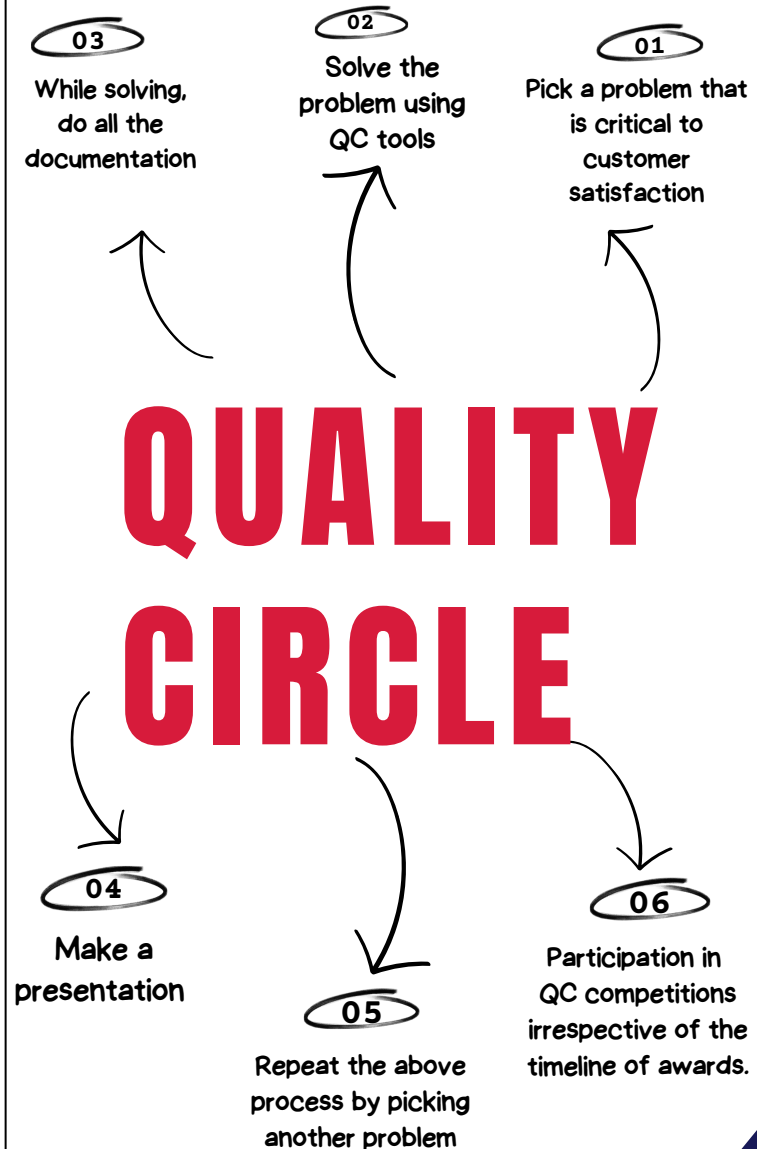
The essence of quality circles lies in the collective effort of employees to identify and solve problems, enhance processes, and ultimately deliver better results. However, when the focus shifts solely to awards and recognition, the essence of quality circles is neglected, and the potential benefits for the organization are compromised.

When awards and recognition become the primary goal, quality circles can inadvertently foster unhealthy competition among teams. Instead of collaborating and sharing best practices, teams may become secretive and protective of their ideas, fearing that sharing knowledge could diminish their chances of winning. This not only hinders the overall progress of the quality circles but also hampers their effectiveness.

When the primary motivation behind participating in a quality circle is to win an award, the quality of the ideas and solutions generated may be compromised. Instead of focusing on genuine improvements that benefit the organization, participants may be more inclined to prioritize ideas that are more attractive and likely to impress external judges. This can lead to a distorted perception of success, where the appearance of improvement outweighs actual progress.

It is time for the organization to reflect on its approach to quality circles. Are we merely participating in them for the sake of winning awards, or are we genuinely committed to driving internal improvement? Let us not lose sight of the true purpose of quality circles and instead focus on creating an environment where they can flourish and contribute to the long-term success of our organization.

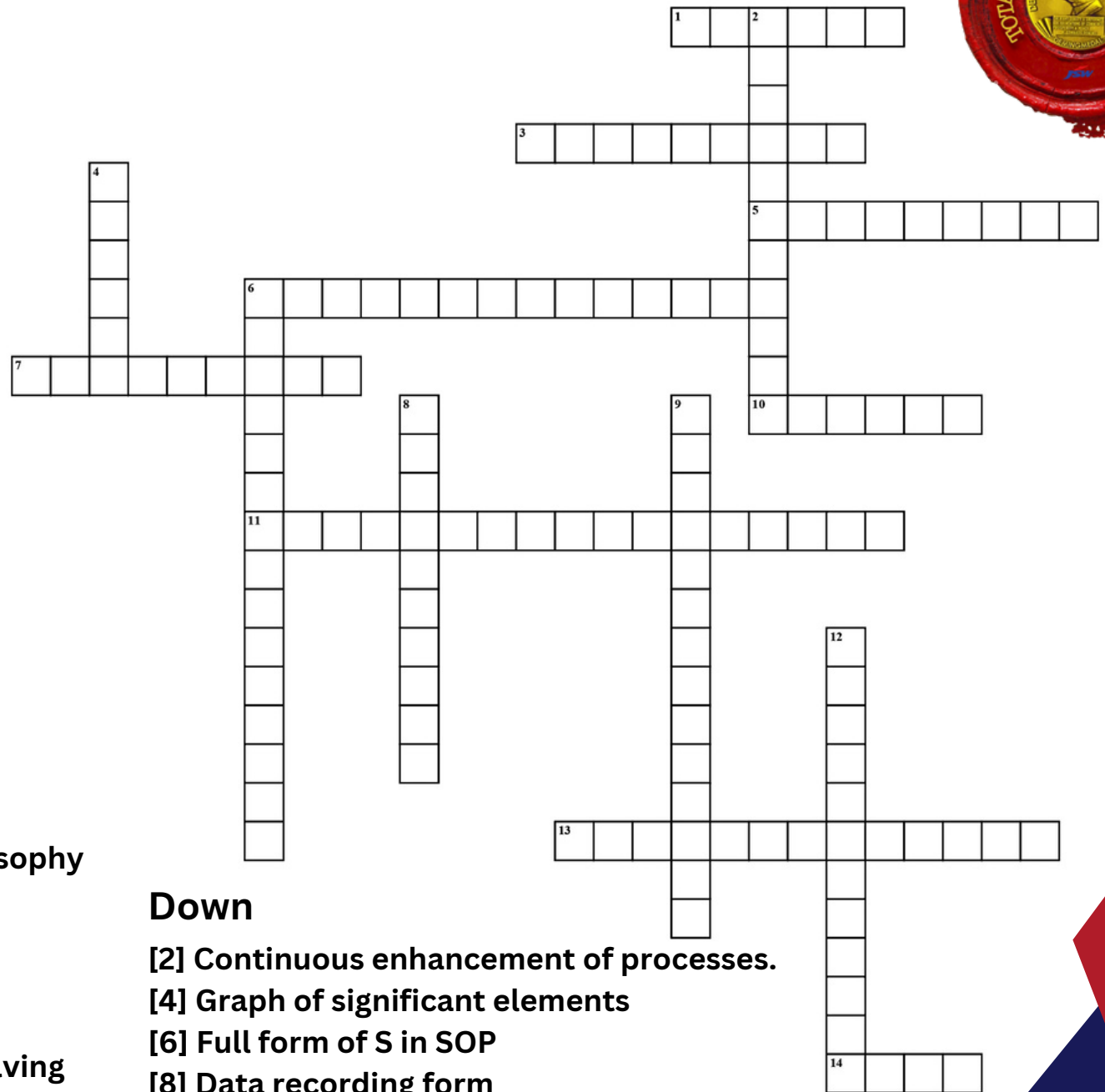
THE DISTORTED PERCEPTION OF





JUMBO CROSSWORD TRIVIA

For answers: follow our coming TQM NEWSLETTER



Across

- [1] Continuous improvement philosophy
- [3] Bar graph displaying data
- [5] Differences in process/data
- [6] Shows variable correlation
- [7] Visual process representation
- [10] No. of Steps in QC problem solving
- [11] Identifies underlying causes
- [13] Creative idea generation
- [14] Group working together

Down

- [2] Continuous enhancement of processes.
- [4] Graph of significant elements
- [6] Full form of S in SOP
- [8] Data recording form
- [9] Gathering relevant information
- [12] Monitors process performance



References:

- 1) Ganapathy K., Narayana V., & Subramaniam B.
Quality Circle Concept and Implementation Book.
Publisher: [QCFI]. 2002.
- 2) TQM.JSW.IN. Accessed [02/09/2023].
Available at: tqm.jsw.in.
- 2) QCFI.IN Accessed [09/09/2023].
Available at: qcfi.in.
- 3) JUSE. Accessed [07/09/2023].
Available at: www.juse.or.jp.

This magazine on quality circle incorporates the essential knowledge and expertise of numerous authors and resources, contributing to its comprehensive nature.

The primary source for this magazine is the "Quality Circle Concept and Implementation Book" authored by K. Ganapathy, Prof. V. Narayana, and B. Subramaniam. This book serves as the foundation for understanding quality circle principles and their practical application.

Additionally, the magazine draws inspiration and insights from the website TQM.JSW.IN, which provides valuable information related to Total Quality Management (TQM). This web resource serves as a contemporary reference to further enhance the understanding of quality circle implementation.

Furthermore, the magazine acknowledges the Japan Union of Scientists and Engineers (JUSE), an organization dedicated to promoting excellence in quality and management practices. The JUSE website (www.juse.or.jp) serves as an additional source of valuable insights and references related to quality circles and overall quality management.

Another key reference incorporated in this magazine is the QCFI portal, a respected platform dedicated to promoting quality circle initiatives and best practices throughout industries. The insights and resources available on the QCFI portal have further enriched the content and context of this publication.

By incorporating these references and acknowledging the contributions of esteemed authors, and web resources, this magazine aims to provide readers with a comprehensive understanding of quality circles and total quality management concepts.

REFERENCES